

Appendix J - Full Memory Map

The fixed low regions of the Intuition Engine 64-bit physical bus, in address order, with their size and purpose. IE64 can address beyond this low window when RAM is present. IE32, M68K, and x86 see the low 4 GB window directly; the 6502 and Z80 see windowed views into it (Chapters 26 and 27).

J.1 Low RAM and stacks

Range	Size	Purpose
\$000000-\$0003FF	1 KB	M68K vector table; x86 IVT; trap vector base for IE64.
\$000400-\$022FFF	~140 KB	BASIC ROM image, system call shim, monitor stub.
\$023000-\$04FFFF	180 KB	BASIC program text (BASIC_PROG_START-).
\$050000-\$057FFF	32 KB	BASIC simple variables.
\$058000-\$05FFFF	32 KB	BASIC string variables.
\$060000-\$08BFFF	176 KB	BASIC arrays.
\$08C000-\$08FFFF	16 KB	BASIC string temporaries.
\$090000-\$096FFF	28 KB	BASIC GOSUB / FOR stack.
\$097000-\$09EFFF	32 KB	Free RAM (general user).
\$09F000-\$09FFFF	4 KB	IE32 stack (STACK_START).

J.2 PC-compatible VRAM apertures

Range	Size	Purpose
\$0A0000-\$0AFFFF	64 KB	VGA graphics window (mode 13h, mode 12h linear).
\$0B0000-\$0B7FFF	32 KB	Reserved (PC-compatible monochrome text window; not implemented).
\$0B8000-\$0BFFFF	32 KB	VGA text buffer (\$B8000).

J.3 General RAM gap

Range	Size	Purpose
\$0C0000-\$0CFFFF	64 KB	Free RAM (general user).
\$0D0000-\$0DFFFF	64 KB	Voodoo texture RAM.
\$0E0000-\$0EFFFF	64 KB	Free RAM (general user).

J.4 The MMIO region (\$F0000-\$FFFFF)

Range	Size	Device
\$F0000-\$F049B	1180 B	Video chip + palette + extended blitter.
\$F0700-\$F07FF	256 B	Terminal / serial / input.
\$F0800-\$F0B7F	896 B	SoundChip channel registers.
\$F0B80-\$F0B91	18 B	AHX player.
\$F0BC0-\$F0BD7	24 B	MOD player.
\$F0BD8-\$F0BF3	28 B	WAV player.
\$F0C00-\$F0C20	33 B	PSG / AY.
\$F0C30-\$F0C3F	16 B	SN76489.
\$F0C40-\$F0CFF	192 B	SoundChip flex channels 4-6.
\$F0D00-\$F0D20	33 B	POKEY.
\$F0D40-\$F0DFF	192 B	SoundChip flex channels 7-9.
\$F0E00-\$F0E1C	29 B	SID primary registers.
\$F0E20-\$F0E2D	14 B	SID player block.
\$F0E30-\$F0E4C	29 B	SID2 registers.
\$F0E50-\$F0E6C	29 B	SID3 registers.
\$F0E80-\$F0EFF	128 B	SFX triggers.
\$F0F00-\$F0F5F	96 B	TED audio + video.
\$F1000-\$F13FF	1 KB	VGA registers.
\$F1400-\$F140F	16 B	HOST appliance block.
\$F2000-\$F2017	24 B	ULA registers.
\$F2100-\$F213F	64 B	ANTIC.
\$F2140-\$F21FB	188 B	GTIA.
\$F2200-\$F221F	32 B	File I/O.
\$F2260-\$F22AF	80 B	Amiga Paula DMA.
\$F2300-\$F231F	32 B	Media loader.
\$F2320-\$F233F	32 B	RUN loader block.
\$F2340-\$F238F	80 B	Coprocessor.
\$F2390-\$F23AF	32 B	Clipboard bridge.
\$F23B0-\$F23BF	16 B	Coprocessor extended monitor.
\$F23C0-\$F23DF	32 B	IRQ diagnostics.
\$F23E0-\$F23FF	32 B	Bootstrap loader.
\$F2400-\$F24FF	256 B	SysInfo (RAM-size ABI).

Range	Size	Device
\$F8000-\$F87FF	2 KB	Voodoo 3D registers.
\$FA000-\$FBAFF	6912 B	ULA VRAM aperture.

J.5 Main video RAM

Range	Size	Purpose
\$100000-\$5FFFFFF	5 MB	Main VRAM aperture for VideoChip framebuffers; large modes may point VIDEO_FB_BASE into ordinary RAM.

J.6 Low-window RAM and reserved aliases

Range	Size	Purpose
\$00600000-\$FFFEFFFF	Dynamic	Extended RAM when backed by the current guest-RAM allocation.
\$FFFF0000-\$FFFFFFFF	64 KB	Sign-extended alias of \$00000000-\$0000FFFF.

Guest RAM can extend beyond these fixed low ranges. Use the SysInfo registers in Chapter 23 to discover total and active visible RAM. IE64 can address backed RAM above \$FFFFFFFF; the compatibility CPUs remain inside the low window or their documented profile caps.

J.7 The 6502 / Z80 views

The 6502 sees a 16-bit address space \$0000-\$FFFF. Its adapter routes the high page-window region as follows:

6502 range	Maps to low bus address
\$0000-\$00FF	zero page in main RAM.
\$0100-\$01FF	6502 stack page in main RAM.
\$0200-\$CFFF	main RAM.
\$D200-\$D20A	POKEY registers.
\$D400-\$D40F	PSG registers.
\$D500-\$D55F	SID registers.
\$D600-\$D605	TED audio registers.
\$D620-\$D632	TED video registers.
\$D700-\$D70D	VGA registers.
\$D800-\$D817	ULA registers, including the paged VRAM data port.
\$E000-\$EFFF	bank-selectable window into main RAM.
\$F000-\$FFF9	MMIO mirror of \$F0000-\$F0FF9, with \$F700-\$F705 and \$F7F0 intercepted as bank registers.
\$FFFA-\$FFFF	NMI / reset / IRQ vectors.

The Z80 sees the same shape with port I/O substituted for the chip-aperture region; the bank-window scheme is identical.

J.8 Notes

The map above is the compact engineering view of the low and device regions that have fixed meanings inside the 64-bit physical bus. Most user programs use a small sub-set of it - the BASIC variable area, the I/O region, and the main video RAM - and never touch the rest. Programs that compete for memory (a large in-memory dataset, a custom blitter routine, a coprocessor worker that wants its own heap) should carve their storage out of the "Free RAM" gaps in J.1, J.3, and J.6, or from boot-sized extended RAM reported by SysInfo, rather than colliding with the BASIC variable region.